Cairo University	Academic Year:	2020/2021	Term Exam:	Second Term	
	Department:	Electronics & Electrical Communications	Course Code:	ELC458	
	Date:	20/6/2021	Course Title:	Selected topics in Digital Integrated Circuits	
	Time:	2 Hours	Full Mark:	70	

1- Write an expression for the clock period and the hold-time condition when there is a negative skew observed between two registers.

2- Explain briefly why you would want to have negative skew instead of positive skew if you had the option.

3- Explain why jitter has no impact on hold time condition

4-Write an expression for the clock period of a pipeline suffering from jitter

5-Write one **disadvantage** of the grid method for clock distribution

Examiners Signature

For Students: Please write using a blue pen only

6-Write one **disadvantage** for the tree method for clock distribution

7- This piece of code includes one mistake. This mistake will not produce an error (neither in the compiler nor the synthesizer). Describe why this mistake is problematic and write down the code with the mistake solved. Draw the logic circuit (high level circuit like decoder, encoder, etc) represented by the fixed code.

Original code	Fixed code

Process(sel)
Begin
  If (sel=”00”) then
  A<=’0’; B<=’1’; C<=’0’;
  Elsif (sel=”01”) then
  A<=’0’; D<=’0’; C<=’0’;
  Elsif (sel=”10”) then
  A<=’1’; B<=’1’; C<’1’;
  Else
  A<=’1’; B<=’0’; C<’1’;
  End if;
End process;	

Why original code is bad (what it would produce):

Logic circuit of fixed code:

Examiners Signature

For Students: Please write using a blue pen only

8- Show all your work for this question. Find the percentage improvement in the critical path delay of an adder when moving from a ripple carry adder to a carry bypass adder architecture. The adder is 40 bits. In the carry-bypass adder, there are 5 blocks each of 8-bits. Use the following:

Tsum=0.69*30RoCo

Tcarry=0.69*18RoCo

Tmux=0.69*4RoCo

Delay of 40-bit ripple carry adder:

Delay of 40-bit carry-bypass adder:

Percentage improvement:

9- Justify why we can pass the read and write pointers in an asynchronous FIFO through synchronizers even though they are busses rather than signals.

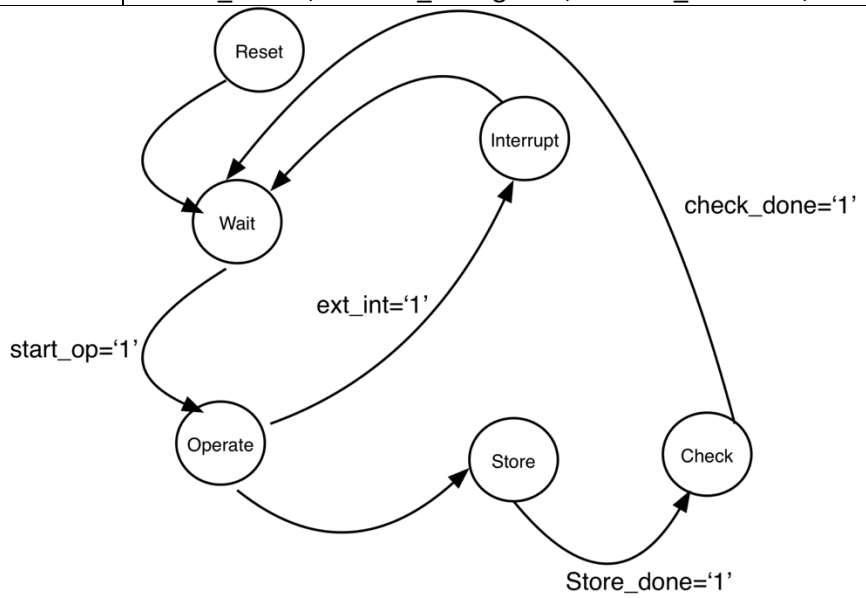
Examiners Signature

For Students: Please write using a blue pen only

10- What is the price you pay for keeping wires thick when scaling technology (wire thickness scaling down slower than width and length).

11- The state diagram below describes a very simple state machine. The table shows the outputs of the state machine for each state.

State	Signals
Reset	Enable_FU='0'; Enable_storage='0'; Enable_tester='0'; Disable_operator='1'
Wait	Enable_FU='0'; Enable_storage='0'; Enable_tester='0'; Disable_operator='0'
Interrupt	Enable_FU='0'; Enable_storage='0'; Enable_tester='0'; Disable_operator='1'
Operate	Enable_FU='1'; Enable_storage='0'; Enable_tester='0'; Disable_operator='0'
Store	Enable_FU='0'; Enable_storage='1'; Enable_tester='0'; Disable_operator='0'
Check	Enable_FU='0'; Enable_storage='1'; Enable_tester='1'; Disable_operator='1'



Draw a block diagram showing the status signals going from the datapath to the state machine and the control signals going from the state machine to the controller.

Examiners Signature

For Students: Please write using a blue pen only

12- Explain why there are no arrows going into the reset state in question 11, but we still can sometimes go to the reset state.

13- Still with question 11, write the code of the next state process of the state machine (the combinational process that determines the next state based on the current state and status signals)

The following type and signals have already been declared for you:

Type state is (reset, wait, interrupt, operate, store, check);

Signal current_state, next_state: state;

Use the following syntax guides:

- Sensitivity list must be complete
- Do not create implicit latches

Process syntax:

Process(*sensitivity list*)

Begin

Body

End process;

Case syntax:

Case *switch* is

When *choice =>*

Code

When *choice2=>*

Code

..

When others=>

Code

End case;

Examiners Signature

For Students: Please write using a blue pen only

14- When moving between two technologies, dimensions scale down by a factor K (L,W,… -> L/K, W/K,…), voltages by a factor U (Vdd, Vth, … -> Vdd/U, Vth/U, …), and the die dimension grows by a factor Kw. Find an expression for the scaling of the resistance of a global wire

15-Find the scaling factor of power dissipation in a global wire (P=IV, V=IR). Use the givens in question 14.

16-In a synchronizer, when can the receiver lower the value of the ack signal. Why can't it lower it at will?

17-Write the expression of critical path delay for the carry-bypass adder in terms of Tmux, Tcarry, Tsum, K (number of bits per block), and P (number of blocks). Find the optimal number of bit bits per block (hint: differentiate)

Expression:

Derivation of optimum:

Examiners Signature

For Students: Please write using a blue pen only

18-If multiplexer delay increases in a carry-bypass adder, would this increase the number of bits per block, or increase the number of blocks? Explain. (You may want to refer to question 17).

19- Show that the time to exit metastability is:

$$T_{exit} = \tau \ln(V_{DD}/V_d(0))$$

Start from the sketch of the two inverters as amplifiers

Examiners Signature

For Students: Please write using a blue pen only

20-Define the following while stressing their differences: Entity, component, instance.

Define entity:

Define component:

Define instance:

Point out the difference between the three:

Examiners Signature

For Students: Please write using a blue pen only